

Q.22 Discuss the significance of maintaining moulds and machines. (CO7)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Describe the injection moulding process in detail, including its applications and advantages. (CO2)

Q.24 Explain the design parameters of moulds, including cooling channels and cycle times. (CO5)

Q.25 Discuss the maintenance and safety procedures for moulds and machines, highlighting best practices. (CO7)

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3th Year / Advance Diploma in Tool & Die Making Subject : Tool Design III (Plastic Moulds)

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which type of plastic is known for its ability to be re-molded? (CO1)

- a) Thermoplastic b) Thermosetting
- c) Elastomer d) Composite

Q.2 What is the main function of the clamping system in a moulding machine? (CO3)

- a) Cooling the mould
- b) Heating the mould
- c) Holding the mould closed during operation
- d) Ejecting the finished part

Q.3 Which moulding process uses a heated cylinder and screw to inject material? (CO5)

- a) Blow Moulding
- b) Compression Moulding
- c) Injection Moulding
- d) Rotation Moulding

- Q.4 What is the primary function of a sprue in a mould? (CO5)
- Cool the material
 - Provide a channel for molten material
 - Eject the finished part
 - Hold the mould together
- Q.5 Which of the following is NOT a parameter to consider when designing a mould? (CO4)
- Depth of cavity
 - Material shrinkage
 - Machine color
 - Ejection system
- Q.6 The process of applying a protective coating to a mould is known as : (CO7)
- Polishing
 - Finishing
 - Surface treatment
 - Moulding

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The materials used for mould construction must have _____ thermal conductivity. (CO1)
- Q.8 The purpose of a cooling system in a mould is to _____ the cycle time. (CO7)
- Q.9 A _____ is used to trim excess material after moulding. (CO5)
- Q.10 Safety measures in moulding processes include using _____ equipment. (CO7)

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- Q.11 The main element that ejects parts from the mould is called the _____. (CO3)
- Q.12 The process of ensuring mould surfaces are smooth is known as _____. (CO5)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the basic components of an injection moulding machine. (CO2)
- Q.14 What are the advantages of using thermoplastics in moulding? (CO1)
- Q.15 Describe the main features of a compression mould. (CO3)
- Q.16 What are the key design parameters for a plastic mould? (CO4)
- Q.17 Explain the blow moulding process and its applications. (CO2)
- Q.18 What considerations must be made for safety in mould operations? (CO7)
- Q.19 How do you prepare an estimate of plastic material requirements for mould making? (CO6)
- Q.20 Describe the procedure for pre-moulding techniques. (CO5)
- Q.21 What is the importance of CAD in mould design? (CO10)

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